L&T EduTech Project Submission

AI-Based Road Surface Defect Monitoring System

Day 3: Ground-Level Model Training & Inference Analysis

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June 2026

3.1 Environment Setup & Hyperparameter Configuration

Day 3 focused on training the first half of our dual-model architecture: the vehicle-mounted camera model. Computations were executed in a Google Colab environment leveraging an NVIDIA Tesla T4 GPU with CUDA 11.8 enabled, ensuring accelerated tensor operations.

We initialized the Ultralytics YOLOv8 architecture, specifically utilizing a pre-trained backbone to leverage transfer learning. The network consists of a CSPDarknet feature extractor and a Path Aggregation Network (PANet) neck, which is crucial for fusing multi-scale features—allowing the model to detect a massive pothole and a hairline crack in the same frame. The hyperparameters were set to a learning rate of 0.01 using the AdamW optimizer, a batch size of 16 to maximize GPU VRAM without causing memory fragmentation, and an input image dimension of 640×640 pixels.

3.2 Training Convergence & Loss Function Analysis

The training process stabilized smoothly over the defined epochs. YOLOv8 utilizes a multi-part loss function: Box Loss (bounding box regression via CIoU), Class Loss (Binary Cross-Entropy for classification), and Distribution Focal Loss (DFL) to fine-tune the box edges. We observed a steady, monotonic decrease across all three loss metrics on the validation set, indicating that the model was learning generalizing features rather than memorizing the training data (overfitting).

3.3 Quantitative Metrics (mAP & Precision-Recall)

The evaluation metrics generated upon training completion were exceptional:

- **Mean Average Precision (mAP@0.5):** The vehicle model achieved a remarkable aggregate mAP@0.5 of **0.920**.
- **Class-Specific Precision:** Smooth roads achieved near-perfect recognition (0.994), followed closely by Severe Potholes (0.964) and Mild Cracks (0.947). Severe Cracks scored slightly lower (0.878), which is mathematically expected as thin, irregular crack structures are heavily penalized by Intersection over Union (IoU) bounding box strictness.
- **F1-Score Analysis:** The overall F1 curve (the harmonic mean of precision and recall) peaked at 0.89 at a confidence threshold of 0.427. This data point is critical; it defines the exact threshold we will program into our edge-deployment router to separate high-confidence saves from low-confidence cloud escalations.

3.4 Confusion Matrix & Error Analysis

An analysis of the generated Confusion Matrices revealed insightful intra-class dynamics. The most prominent misclassifications occurred exclusively between the *Mild* and *Severe* tiers of the same defect class (e.g., predicting D00 when the ground truth was D10). This error mode is physically acceptable; defect degradation is a continuous spectrum, and the boundary between "mild" and "severe" often relies on subjective pixel depths that 2D cameras struggle to perfectly quantify.

Crucially, background false positives were practically non-existent. The model successfully ignored complex background textures like varying asphalt aggregate and shadows.

3.5 Qualitative Validation & Export

Testing inference on held-out validation batches demonstrated excellent scale invariance. Visual inspection of the `val_batch_pred` outputs showed the network successfully placing tight bounding boxes around localized severe potholes in the foreground while simultaneously detecting minor longitudinal cracks near the vanishing point of the road. The final optimized weights (`best_road.pt`) were successfully exported in PyTorch format, ready for the upcoming dashboard integration.